EE1007
Introduction to Computer Science Laboratory

Lab 10
Pointer (I)

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Lab 10.1
Reverse A List

- In this problem, you are asked to write a program for reversing a given list.
- The program will take a list which consists of several integers as the input, and output a list in the reversed order.
- The programmer **must** use **reverse function** to reserve the list. The reverse function must contain the **swap function**. The function prototype are shown below:
  - **void reverse(…);**
  - **void swap(long long *const, long long *const);**
- Moreover, **cout or printf** can only be used in main function but no other functions.
- Violating the above two rules will result in 0 point of this problem.
- The maximum number count in the list is 50, and the range of each input integer is $[-2^{63} \sim 2^{63}-1]$

Lab 10.1
Reverse A List

- The screen dialog should appear as follows:
  - First line shows the input sequence provided by the user, and the second line shows the output of the program.
- Example 1 (1 2 3 4 5 6 7 8 9)

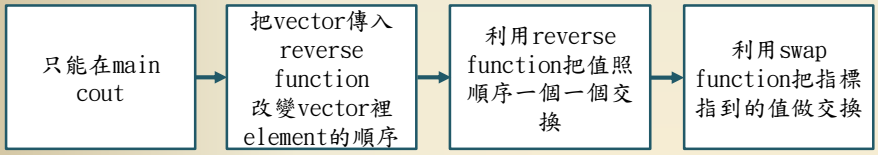
```
1 2 3 4 5 6 7 8 9
9 8 7 6 5 4 3 2 1
```

- Example 2 (10 30 20 40 60)

```
10 30 20 40 60
60 40 20 30 10
```


Lab 10.1
Reverse A List

- 思考方向:



Lab 10.1
Reverse A List

- How to get input list:

```
int i=0, num;
int list[50];
while(cin >> num){
    list[i] = num;
    if(cin.get() == '\n') break;
    i++;
}
```

- How to use pointer to change value:

```
void swap(long long *const num1, long long *const num2){
    long long temp;
    temp = *num1;
    *num1 = *num2;
    *num2 = temp;
}
```


Lab 10.2
Find Mountains

- In this problem, you are asked to write a program to **find the longest mountain** and **the number of the mountains** contains in a given list contains several integers as elements.
- The elements in a list represent the altitudes. A sequence of elements forms a hill is called mountain. For example, a sequence [1 4 6 5 4] forms a mountain.
- A list may consist of several mountains. For example, a list =[1 4 6 5 4 7 8 5] as shown in the figure contains 2 mountains: [1 4 6 5 4] and [4 7 8 5].
- The longest mountain of the list is defined as the mountain contains maximum elements. For example, [1 4 6 5 4] is the longest mountain of the list =[1 4 6 5 4 7 8 5].



Lab 10.2
Find Mountains

- The program will take a list from standard input, and show the number of mountains as well as the longest mountain.
- The programmer **must** use the function prototypes shown below to find the number of mountains as well as the longest mountain:
- **void find_mountain(int *const num, int *const Length, …);**
  - **num** represents the number of the mountains
  - **Length** represents the longest length of the mountain
- Moreover, **cout or printf** can only be used in main function but no other functions.
- Violating the above two rules will result in 0 point of this problem.
- The maximum number count in the list is 50, and the range of each input integer is $[-2^{63} \sim 2^{63}-1]$.

Lab 10.2
Find Mountains

- The screen dialog should appear as follows:
  - First line shows the input sequence provided by the user. The second and third line show the output of the program where the second line is the number of mountains and the third line is the longest mountain.
- Example 1 (1 4 6 5 4 7 8 5)

```
1 4 6 5 4 7 8 5
2
1 4 6 5 4
```

- Example 2 (10 11 13 12 5 10 12 13 12 11 50 100)

```
10 11 12 5 10 12 13 12 11 50 100
2
5 10 12 13 12 11
```

Lab 10.2
Find Mountains

◆計算山的數目的思考方向:

只能在main cout → 利用num來儲存山的數目 → 在數值上升後下降，代表有一個山

◆得出最長的山的思考方向:

只能在main cout → 把vector傳到find_mountain function → 紀錄下降完上升的那一個element的位置，當作開頭。把最長的山的長度存在Length裡面

Lab 10.2
Find Mountains

◆How to get the number of the mountain:

在每一次的下降後上升，代表一座山的結束
就可以把山的數目往上加

```
while (i < list.size()) {
    if (list[i] > list[i - 1]) { //increase
        ...
        if (increase after decrease)
            *num += 1;
    }
    if (list[i] < list[i - 1]) { //decrease
        ...
    }
    if (i == (list.size() - 1)) { //a mountain at the end of the list
        ...
        *num += 1;
    }
    i++;
}
```



Lab 10.2
Find Mountains

- How to get the longest mountain:

在每一次的下降後上升，代表一座山的結束
結束以後就可以更新最新的山

```
if (list[i] > list[i - 1]) { //increase
    ...
    if (increase after decrease){
        if ((i - L_temp) > *Length) {
            *Length = i - L_temp;
            L_start = L_temp;
        }
        L_temp = i - 1;
    }
}
```



Lab 10.3
NCU Theater Reservation System(II)

- As more and more student interested in NCU theater, students want to become a member of the NCU theater in order to get lasted information and promotions.
- Based on the previous system you had implemented in Lab 9.3, we would like to add some new features in our NCU theater reservation system.
- In this problem, you are asked to develop a member system can integrated it into the NCU theater reservation system.

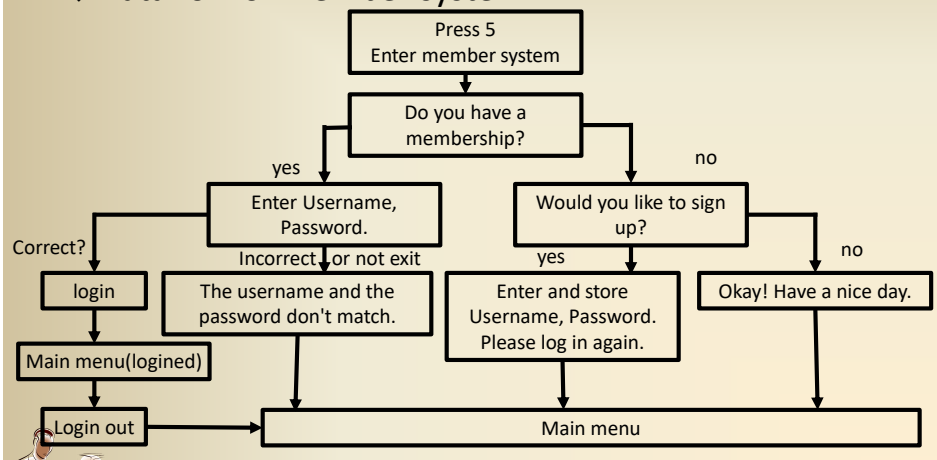
Lab 10.3
NCU Theater Reservation System(II)

- The new main menu of your system is shown as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to exit the system
Enter your selection Here: 5
```


Lab 10.3
NCU Theater Reservation System(II)

- Dataflow of member system



Lab 10.3

NCU Theater Reservation System(II)

- Once user enters 5, the program will go to NCU member system.
- First, the program will check whether the user is a member or not.
- If the user is a NCU member, the program should check whether the user name and password user typed in matches the information the system hold.
  - If the data matched, program should enter the member system. Member system should contain all the functions the normal mode provides.
  - If the data doesn't match, the program should print " **This account doesn't exist**." on the screen.



Lab 10.3

NCU Theater Reservation System(II)

- If the user answers that he is **not** the NCU member, the program should check whether the user wants to enroll a membership.
  - If the user answers "no", program should print **"Okay! Have a nice day."** and quit the member system.
  - If the user answers "yes", program will ask the user to enter the user name and password.
  - The maximum number of the member is 15, and the username/password will contains no more than 50 character
- After user logins the NCU member system, the system should show **"***welcome back *Username****".** Moreover, the system should not show the **"Please type 5 to member center"** in the main menu with login status.
- After enter the NCU member system, if user press 7, the program will log out the NCU member system.



Lab 10.3
NCU Theater Reservation System(II)

- Requirement:
  - If the user enters a wrong input, show the message **"Invalid input, please enter your selection again:"**
- Notice that the program should work at normal mode when there is no one enter the member system.
- The programmer **must** use the function prototypes shown below to deal with the function of member center system, otherwise you will get 0 point.
- **void Member(…);**

Lab 10.3
NCU Theater Reservation System(II)

- The following example of screen dialog are **independent**.
- The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to exit the system
Enter your selection Here: 5

***NCU Theater Member Center***
Do you have the membership? (y/n) y
Enter Username: Andy
Enter Password: 0000
The username and the password don't match.
```

Lab 10.3
NCU Theater Reservation System(II)

◆The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to exit the system
Enter your selection Here: 5

***NCU Theater Member Center***
Do you have the membership? (y/n) n
Would you like to sign up? (y/n) y
Enter Username: Andy
Enter Password: 0000
Welcome! You are now a member of NCU theater.
Please log in again.
```


Lab 10.3
NCU Theater Reservation System(II)

◆The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to exit the system
Enter your selection Here: 5

***NCU Theater Member Center***
Do you have the membership? (y/n) y
Enter Username: Andy
Enter Password: 0000

*** Welcome to NCU theater reservation system ***
***Welcome back Andy***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 6 to exit the system
Please type 7 to log-out
Enter your selection Here:
```

Lab 10.3
NCU Theater Reservation System(II)

- The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to exit the system
Enter your selection Here: 5

***NCU Theater Member Center***
Do you have the membership? (y/n) n
Would you like to sign up? (y/n) n
Okay! Have a nice day.

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to exit the system
Enter your selection Here:
```


Lab 10.3
NCU Theater Reservation System(II)

- Note that there must have one newline between each system.

```
***NCU Theater Member Center***
Do you have the membership? (y/n) n
Would you like to sign up? (y/n) y
Enter Username: Andy
Enter Password: 0000
Welcome! You are now a member of NCU theater.
Please log in again.

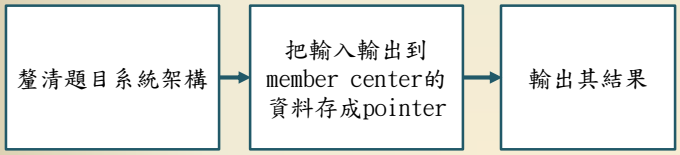
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to exit the system
Enter your selection Here: 5

***NCU Theater Member Center***
Do you have the membership? (y/n) y
Enter Username: Andy
Enter Password: 0000

*** Welcome to NCU theater reservation system ***
***Welcome back Andy***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
```

Lab 10.3
NCU Theater Reservation System(II)

◆思考方向:



Lab 10.3
NCU Theater Reservation System(II)

◆When user answer that they don't have membership:

```
cout << "***NCU Theater Member Center***" << endl;
cout << "Do you have the membership? (y/n) ";
cin >> choice;

while (choice != 'y' && choice != 'n') {
    cout << "Invalid input, please enter your selection again: ";
    cin >> choice;
}

if (choice == 'y') {
    cout << "Enter Username: " ;
    cin >> username;
    cout << "Enter Password: " ;
    cin >> password;

    for (int i = 0; i < 15; i++) {
        if (!strcmp(username, member_name[i])) {
            if (password == member_data[i][0]) {
                *login = i;
                cout << endl;
            }
        }
    }

    if (*login == -1) {
        cout << "The username and the password don't match." << endl << endl;
        return;
    }
}
```

Lab 10.3
NCU Theater Reservation System(II)

- When user answer that they have membership :

```
else if (choice == 'n') {
    cout << "Would you like to sign up? (y/n) ";
    cin >> sign_up;

    while (sign_up != 'y' && sign_up != 'n') {
        cout << "Invalid input, please enter your selection again: ";
        cin >> sign_up;
    }

    if (sign_up == 'y') {
        for (int i = 0; i < 15; i++) {
            if (member_data[i][0] == -1) {
                user_id = i;
                break;
            }
        }

        cout << "Enter Username: " ;
        cin >> member_name[user_id];
        cout << "Enter Password: " ;
        cin >> member_data[user_id][0];
    } else if (sign_up == 'n') {
        cout << "Okay! Have a nice day." << endl << endl;
        return;
    }

    cout << "Welcome! You are now a member of NCU theater." << endl
         << "Please log in again." << endl << endl;
}
```


Lab 10.4
Inserting + and *

- In this problem, you are asked to write a program for inserting + or * into a given list to fit the computation result to a given number.
- The program will take two inputs: a list of numbers and another single number. Then the program need to find out all possible combinations of inserting + and * to the given list so that the computation result will fit the single number. Then the program will print all possible results on the screen.
- You should follow the arithmetic precedence.
- If it's not possible to insert + and * to match the target, the program should show **"not found".**
- Moreover, the programmer **must** use the function prototypes shown below to deal with the function of swapping element of the list.
- **void insertOp(…);**
- Note that the program can use **cout or printf** only in the main function.
- Violating the above two rules will result in 0 point of this problem.

Lab 10.4
Inserting + and *

- The output of the possible ways to insert + or * should start with the + and then *. For example, the result should be [1+2*1], [1*2+1] instead of [1*2+1], [1+2*1].
- Notes that the sequence of the list can't be changed. For example, the input[1 2 3] can't be seen as [1 3 2], [2 1 3], etc.
- The maximum number count in the list is 50, and the range of each input integer is [0~2^{63}-1].

Lab 10.4
Inserting + and *

- The screen dialog should appear as follows:
  - First line shows the input sequence provided by the user. Second line is goal number we want to achieve. Third line shows all the possibility to insert + and *.
- Example 1 (1 2 3)

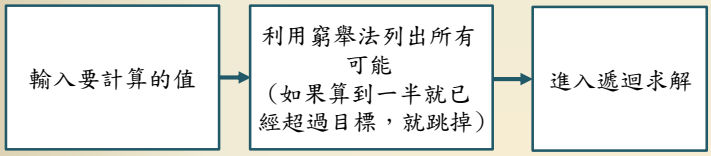
```
1 2 3
6
[1 + 2 + 3], [1 * 2 * 3]
```

- Example 2 (1 20 30 50 60)

```
1 20 30 50 60
2
not found
```

Lab 10.4
Inserting + and *

◆ The screen dialog should appear as follows:



Lab 10.4
Inserting + and *

◆ Exhaustive method :

```
void insertOp(vector<vector<int>> &result, vector<int> list, vector<int> temp, int num, int prev) {
    int calculated;
    if (list.size() == 1) {
        if (num == (target number)) {
            result[result_num++]=temp;
            temp.clear();
            return;
        } else {
            temp.clear();
            return;
        }
    } else if (num > (target number)) {
        temp.clear();
        return;
    } else {
        list.assign(list.begin() + 1, list.end());
        //add+
        calculated = num + list[0];
        temp[index++] = -1;
        temp[index++] = list[0];
        insertOp(result, list, temp, calculated, previous_num, l_done);
        //add*:
        calculated = (num - prev) + prev * list[0];
        temp[index++] = -2;
        temp[index++] = list[0];
        insertOp(result, list, temp, calculated, previous_num, l_done);
    }
}
```

最小的可以解決的問題 => 當看到最後一個element

把問題縮小 => 把list縮小

Insert +

Insert *

由於如果需要做insert*就必須還原上一個+，所以需要一個previous number來還原
For example:
[1, 2, 3] target=6
1 num=1
1+2 num=1+2=3
1+2*3 num=(3-2) + 2*3=6
所以這邊最後一個操作的previous number就會是2



Evaluation



- Lab10.1(30%)
  - 5 testcases, 6% each
- Lab10.2(30%)
  - 5 testcases, 6% each
- Lab10.3(30%)
  - 5 testcases, 6% each
- Lab10.4(20%)
  - 5 testcases, 4% each

